



Alberta Public Health Disease Management Guidelines

Campylobacteriosis



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Health and Wellness Promotion Branch

Public Health and Compliance Branch

Alberta Health

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Case Definition

Confirmed Case

Laboratory confirmation of infection with or without clinical illness^(A):

- Isolation of *Campylobacter* species from an appropriate clinical specimen (e.g., stool and blood)

Probable Case (Outbreak Only)

Clinical illness^(A) in a person who is epidemiologically linked to a confirmed case

^(A) Clinical illness is characterized by diarrhea, abdominal pain, malaise, fever, nausea and/or vomiting.

Reporting Requirements

Physicians, Health Practitioners and Others

Physicians, health practitioners and others shall notify the Medical Officer of Health (MOH) (or designate) of the zone, of all confirmed cases in the prescribed form by mail, fax or electronic transfer within 48 hours (two business days).

Laboratories

All laboratories shall report all positive laboratory results by mail, fax or electronic transfer within 48 hours (two business days) to the:

- Chief Medical Officer of Health (CMOH) (or designate), and
- MOH (or designate) of the zone.

Alberta Health Services and First Nations and Inuit Health Branch

- The MOH (or designate) of the zone where the case currently resides shall forward the initial Notifiable Disease Report (NDR) of all confirmed cases to the CMOH (or designate) within two weeks of notification and the final NDR (amendment) within four weeks of notification.
- The MOH (or designate) shall report all outbreaks to CMOH (or designate) via the [Alberta Outbreak Report Form \(AORF\)](#) as soon as possible using existing processes (e.g., CDOM or fax).
- For out-of-province and out-of-country reports, the following information should be forwarded to the CMOH (or designate) by phone, fax or electronic transfer within 48 hours (two business days):
 - name,
 - date of birth,
 - out-of-province health care number,
 - out-of-province address and phone number,
 - positive laboratory report, and
 - other relevant clinical/epidemiological information.

Epidemiology

Etiology

Campylobacteriosis is an acute zoonotic bacterial infection that can cause both enteric and extraintestinal infections. There are 21 species within the genus *Campylobacter*, but *C. jejuni* and *C. coli* mainly cause diarrhea in humans. *Campylobacter fetus* largely causes systemic illness in neonates and debilitated hosts. Other *Campylobacter* species, including *C. upsaliensis*, *C. lari*, *C. hyointestinalis* can also cause enteric or extraintestinal illnesses.^(1,2)

Campylobacter bacteria can survive in milk, other foods or in water that is kept at 4°C for several weeks. Pasteurization effectively destroys the bacteria, as does appropriate chlorination used for water disinfection.⁽³⁾ *Campylobacter* bacteria also remains viable after freezing.^(4–6)

Clinical Presentation

Infection with *Campylobacter* may present with variable severity of symptoms. Predominant symptoms include diarrhea (may contain visible or occult blood), malaise, fever, abdominal pain and sometimes vomiting.^(1,2) There may be a prodromal period with fever, headache, myalgia, and general malaise 12–24 hours before the intestinal symptoms appear.⁽⁷⁾ Some infected individuals may be asymptomatic. It will depend on the dose of the infective organism reaching the small intestine, the virulence of the infecting strain and the specific immunity of the exposed person.⁽³⁾ Bacteremia is uncommon but can occur in children.⁽¹⁾

Campylobacteriosis is generally self-limited and the majority of individuals fully recover. Symptoms typically resolve in less than one week; however, in about 10–20% of cases, relapses or prolonged or severe illness can occur.⁽¹⁾

Post infection complications are rare but can include such things as Guillain-Barré syndrome (GBS), reactive arthritis/Reiter's syndrome, Miller Fisher syndrome, myocarditis, pericarditis, urticaria and erythema nodosum.^(1,2) Case fatality rates have been estimated to range from 0.01 to 1%.⁽²⁾

Diagnosis

Diagnosis is made by culture of the organism from an appropriate clinical specimen (e.g., stool or blood).

Treatment

- In most cases, infection is self-limited and treatment with antibiotics is not indicated. Rehydration and electrolyte replacement are considered the primary treatment.⁽²⁾
- Antibiotics, if required, shorten the duration of illness and excretion of organisms and prevent relapse when given early in gastrointestinal tract infection.⁽¹⁾
- Treatment with appropriate antibiotics (e.g., azithromycin, erythromycin) will usually eradicate the organism from the stool within two to three days.⁽¹⁾

Reservoir

The gastrointestinal tracts of domestic and wild birds and animals are reservoirs of *Campylobacter* infections.⁽¹⁾ Animals most commonly affected include poultry and cattle, but other potential sources of infection include puppies, kittens, other domestic animals, swine, sheep, rodents and all types of fowl.^(2,3) *Campylobacter jejuni* and *C. coli* have been isolated from feces in 30–100% of healthy chickens, turkeys and water fowl.⁽¹⁾ Raw poultry or meat, often contaminated through the slaughter process, and unpasteurized milk have been frequently identified as sources of infection. Soil and water can be contaminated by excreta from infected animals.⁽³⁾

Transmission

The most common modes of transmitting *Campylobacter* bacteria include:

- ingesting contaminated food (particularly raw or undercooked poultry or meat),
- drinking unpasteurized milk or contaminated/untreated water, and
- having close contact with fecal material from infected animals and pets or infected people.

It has been estimated that 50–70% of sporadic *Campylobacter* infections in developed countries are as a result of consuming undercooked poultry.⁽³⁾ Cross-contamination from cutting boards may cause infections, especially when raw poultry is cut on them.⁽²⁾ Although person-to-person transmission is uncommon, it may occur and the risk is greatest during the acute phase of the illness.⁽¹⁾ Newborns of infected mothers may become infected. The infective dose for *Campylobacter* is considered to be low (as few as 500 organisms).⁽²⁾

Incubation Period

Usually two to five days, but can range from one to 10 days depending on the dose ingested.⁽²⁾

Period of Communicability

Campylobacter is communicable throughout the course of the infection typically lasting several days to several weeks. Persons not treated with antibiotics may excrete the organism for as long as two to seven weeks.⁽²⁾

Host Susceptibility

The immune mechanisms are not well understood but lasting immunity to serologically related strains follows infection. In developing countries, the majority of the population will develop immunity in the first two years of life.⁽²⁾ Immunocompromised hosts have an increased risk for prolonged, relapsing or extraintestinal infections.⁽¹⁾ Decreased stomach acidity has also been reported as a risk for infection.⁽²⁾

Incidence

General

Campylobacter is a leading cause of bacterial diarrheal disease worldwide, and it is an important cause of traveller's diarrhea.⁽⁸⁾ Common source outbreaks have occurred, most often associated with foods.⁽²⁾

Canada

In 2012, *Campylobacter* infections accounted for about 43% of all reported cases of enteric, food and waterborne diseases in Canada (for both sexes, including unknown, and for all ages).⁽⁹⁾ The rates for *Campylobacter* infections in Canada since 2000 have been generally decreasing with a slight upward trend in 2009–2012. The rates ranged from 40.66 per 100,000 (2000) to 29.3 per 100,000 in 2012.⁽⁹⁾

Alberta

The overall trend of rates in Alberta between 2000 and 2014 were similar to those in Canada as a whole, and also showed a general downward trend. The peak was in 2002 with a rate of 45 cases per 100,000 (1390 cases) and has been steady since 2007, with a rate between 23 and 26 per 100,000.⁽¹⁰⁾

Refer to the [Interactive Health Data Application \(IHDA\)](#) for more information.

Public Health Management

Key Investigation

- Confirm that the case meets the case definition.
- Obtain a history of illness including the date of onset, signs and symptoms.
- Identify any underlying medical conditions that may increase host susceptibility.
- Determine the occupation of the case and if in a **sensitive situation or occupation (SSO)** that poses a higher risk of transmission to others. See Table 1 for a definition of SSO.
- Determine the possible source of infection, taking into consideration the incubation period^(B), reservoir, and mode of transmission. Assessment may include determining, obtaining or identifying:
 - a detailed food history including recent consumption of potentially contaminated food (especially poultry, beef, pork), contaminated water, or unpasteurized milk,
 - possible sources of cross contamination (e.g., cutting boards),
 - possible exposure to domestic or wild animals or fowl including identifying recent illness in pets or acquisition of a puppy, kitten, etc. into the household,
 - occupational exposure (e.g., animal or meat handling),
 - high-risk sexual practices, especially contact with feces,
 - recent travel or immigration, and/or
 - residing in areas with poor sanitation including improper water treatment and sewage disposal either in Canada, or abroad.
- Obtain implicated food samples, if possible.⁽²⁾
- Assess for history of similar symptoms in other members of the household.
- Identify contacts that may have had significant risk of direct or indirect exposure to the feces of the case, especially those in SSO (refer to Table 1). Contacts include:
 - persons living in the household,
 - children and child care workers at a child care facility (daycare, dayhome, or other child care site), and
 - individuals exposed to the same source (if identified).

Table 1: Sensitive Situations or Occupations (SSO)⁽¹⁾

SSO	Activities
Food handler	• Touches unwrapped food to be consumed, <u>and/or</u> • Handles equipment or utensils that touch unwrapped food to be consumed.*
Health care, child care or other staff	• Has contact through serving food to <u>highly susceptible</u> patients or persons in whom an intestinal infection would have particularly serious consequences. • Provides direct patient care and is involved in the care of young children, elderly or dependent persons.
Child attending a child care facility or similar facilities	• Is diapered or unable to implement good standards of personal hygiene.
Any individual (older child or adult)	• Is unable to implement good standards of personal hygiene (e.g., with disabilities/challenges that may impact ability to perform good hand hygiene and is involved in an activity that may promote disease transmission).

* NOTE: Generally, food handlers who do not touch food, equipment or utensils in this way are not considered to pose a transmission risk; however, circumstances for each case should be assessed on an individual basis.

^(B) The incubation period of campylobacteriosis is usually 2–5 days; however, individual cases can vary requiring flexibility when determining where and how the disease was likely acquired.

Management of a Case

- All cases should be advised of the following:
 - how the disease is transmitted, appropriate personal hygiene, routine infection prevention and control practices, and contact precautions, and
 - to avoid food preparation until symptoms have resolved.
- For hospitalized children or adults unable to maintain appropriate hygiene habits or who have incontinence that cannot be contained, additional precautions (i.e., contact precautions) should be implemented.⁽¹¹⁾ Consultation with facility Infection Control staff would be appropriate.
- Refer to Table 2 for case exclusion criteria. For a summary of all enteric exclusions refer to [Enteric Transmission Risk Assessment and Exclusion Table](#).

Table 2: Case Exclusion

Cases	Category	Exclusion Criteria
Symptomatic	SSO	• The MOH may by order exclude the case until 48 hours after appropriate antibiotic treatment has been completed and stools have returned to normal or the MOH is satisfied that the case is no longer infectious. – The case must be symptom free for 48 hours after stopping any antidiarrheal medication (if taken). – Lifting of exclusions is not conditional upon submission of stool specimens to demonstrate clearance of the organism, and specimens may still be submitted on a case-by-case basis in consultation with the MOH. – Specimens may still be submitted on a case-by-case basis in consultation with the MOH. • If possible, consideration may be given to temporary redeployment away from activities that involve increased risk of transmission.
Asymptomatic	SSO	• Generally not required unless otherwise recommended by the MOH.
Symptomatic	Non-SSO	• No exclusion required; however, all cases of gastroenteritis or enteritis should be regarded as potentially infectious and should remain home from work, school or daycare until 48 hours after diarrhea has stopped.
Asymptomatic	Non-SSO	• No exclusion required.

Management of Contacts

- Contacts should be instructed about disease transmission, appropriate personal hygiene, routine practices, and contact precautions.
- Symptomatic contacts should be assessed by a physician.
 - Contacts with positive stool specimens should be treated as cases.
- Refer to Table 3 for contact exclusion criteria.

Table 3: Contact Exclusion

Contacts	Category	Exclusion Criteria
Symptomatic	SSO	• Same as a case. • Ensure the contact is assessed by their physician.
Symptomatic	Non-SSO	• No exclusion required however, all cases of gastroenteritis or enteritis should be regarded as potentially infectious and should remain home from work, school or daycare until 48 hours after diarrhea has stopped. • Refer to their physician for assessment and testing, if required.
Asymptomatic	All	• No exclusion – contacts should monitor themselves for gastrointestinal symptoms, maintain good hand hygiene and food handling practices and seek medical attention if symptoms develop.

Preventive Measures

- Educate the **public** about the following:^(1,2)
 - exercising good hand hygiene to prevent cross-contamination with other foods after handling raw poultry,
 - washing cutting boards, counter tops and utensils with soap and water after contact with raw meat,
 - cooking meats thoroughly,
 - washing hands after contact with farm animals, pets, animal feces, and animal environments, especially where the animals/pets are ill with diarrhea,
 - consuming only pasteurized milk, dairy products, and juices,
 - accessing and drinking safe water supplies, and
 - testing private water supplies for presence of bacterial contamination, if suspected
- Educate **food handlers** about:^(1,2)
 - proper food and equipment handling, preparation, and hygiene, especially in avoiding cross-contamination from raw meat products, and
 - thorough hand washing before and after meal preparation.

Appendix 1: Revision History

Revision Date	Document Section	Description of Revision
November 2021	General	• Updated Template • Etiology, Clinical Presentation, Diagnosis and Treatment sections moved to Epidemiology • Key Investigation section moved to Public Health Management (formerly called Control) • Updated web links
	Case Definition	• Added “(Outbreak Only)” to Probable Case as it is only notifiable during an outbreak.
	Reporting	• Added bullet on outbreak reporting
	Management of Case and Contacts	• Specified where terminology is SSO and included information in tables • Updated Exclusions to align with decisions agreed to in Enteric Transmission Risk Assessment and Exclusion Table • Aligned wording with other enteric diseases
	Preventive Measures	• Reworded to align with other enteric diseases

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